Mobile Sales Performance: Strategic Intelligence Dashboard

1. Situation (The Problem)

In a competitive retail electronics market, stakeholders struggled with a fragmented view of sales performance across different cities and brands. Key operational pain points included:

- **Data Consolidation Issues:** Raw sales records from multiple sources made it difficult to get a unified view of total revenue ($769M) and transaction counts.
- **Static Reporting:** Leadership relied on static reports that couldn't easily filter data by city, mobile model, or payment method in real-time.
- **Lack of Comparative Analysis:** There was no simple way to compare current sales performance ("Month-to-Date") against the same period in the previous year.

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2. Task (The Goals)

The objective was to develop a high-performance, interactive Power BI solution that would:

- **Unify Sales Data:** Create a "Single Source of Truth" for mobile sales across diverse regions (Ludhiana, Delhi, Mumbai, etc.).
- **Enable Dynamic Filtering:** Allow users to slice data by Brand (Apple, Samsung, etc.), Payment Method (UPI, Debit Card), and Time.
- **Analyze Time-Based Trends:** Use advanced calculations to track Month-to-Date (MTD) sales and Year-over-Year (YoY) comparisons.

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3. Action (What Actions Are Performed)

I implemented an end-to-end Business Intelligence workflow within Power BI:

- **ETL & Data Transformation (Power Query):** Imported raw data from Excel/CSV and performed cleaning steps to handle data types and null values for accurate reporting.
- **Relational Data Modeling:** Designed a robust star schema by creating a **Custom Calendar Table** and establishing relationships between the sales data and time dimensions.
- **Advanced Analytics (DAX):** Authored complex **Data Analysis Expressions (DAX)** to calculate critical business metrics, such as:
    - **MTD (Month-to-Date):** To track cumulative sales within a specific month.
    - **Same Period Last Year:** To measure growth and seasonal performance.
- **Interactive Visualization Design:**
    - **Geospatial Analysis:** Integrated a Map visual to highlight sales density by city.

- **Cross-Filtering Interface:** Built a 3-page navigation system (Main Dashboard, MTD Report, and YoY Comparison) with synced slicers for a seamless user experience.

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4. Result (The Outcome)

The final dashboard provided immediate strategic value for the organization:

- **Instant Revenue Visibility:** Management gained real-time access to key metrics: **$769M in Total Sales**, **19K in Quantity**, and **$40.11K Average Price**.
- **Optimized Payment Strategies:** The "Transactions by Payment Method" analysis identified that **Debit Card (26.36%)** and **UPI (24.72%)** were the leading payment drivers, helping focus financial partnerships.
- **Strategic Market Insights:** By analyzing "Total Sales by City," the hospital/company could pinpoint high-performing hubs like **Delhi and Mumbai** for targeted marketing.
- **Performance Benchmarking:** The YoY charts allowed leadership to identify that current sales were outperforming the previous year's figures (e.g., Q1 growth comparison), supporting data-driven expansion plans.